Panduan Hafalan Talk Singkat

From Recorded Flow to Faster Thermal Simulation – cover + 7 slide, sekitar 6 menit

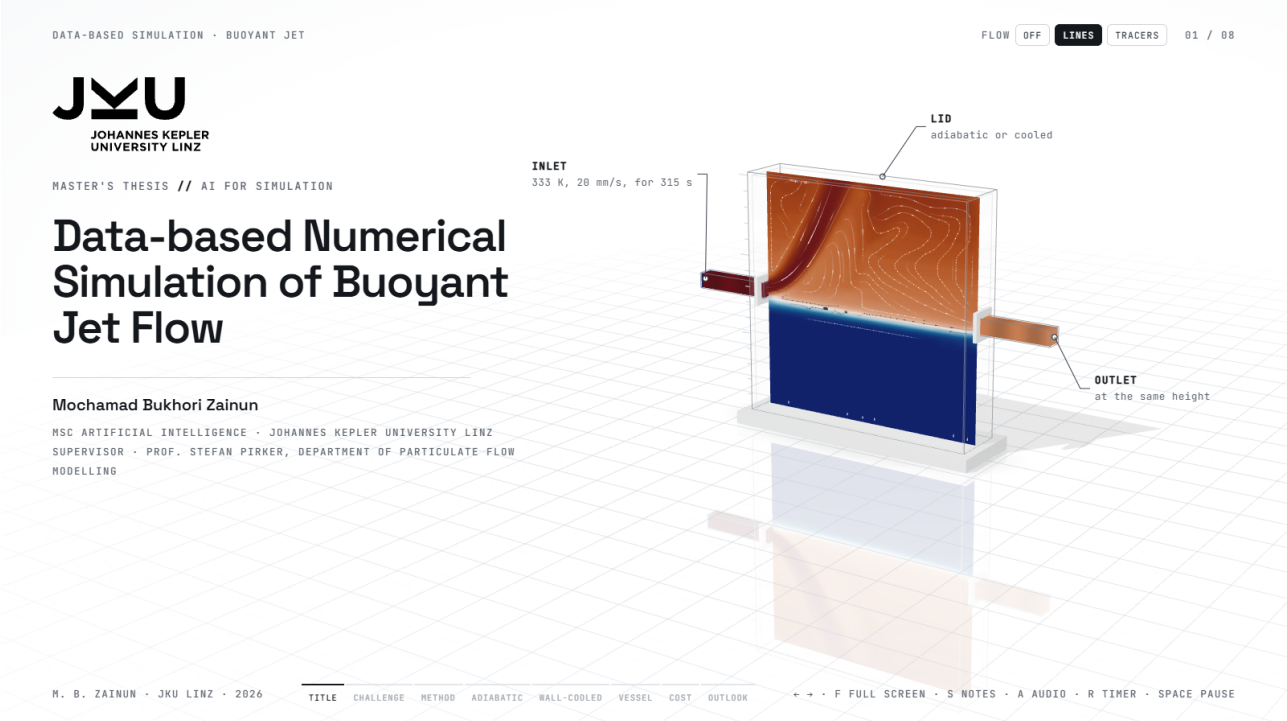
Mochamad Bukhori Zainun – Johannes Kepler University Linz

Cara memakai

1. Lihat gambar slide dulu, lalu baca kotak Alur cerita. Gambar dan alur adalah gantungan hafalan.
2. Ucapkan naskah per kalimat. Garis miring (/) adalah jeda napas.
3. Jangan menghafal kata per kata. Ingat urutan kata kunci, lalu ceritakan dengan kalimat sendiri.
4. Total 824 kata, sekitar 6 menit dengan jeda. Sisa waktu untuk diskusi.

Pertemuan pertama bersifat santai. Tujuannya saling kenal dan mendengar kebutuhan mereka. Kalimat terakhir slide 8 adalah pembuka diskusi.

Slide 1 – Data-based Numerical Simulation of Buoyant Jet Flow *0:30*



Alur cerita

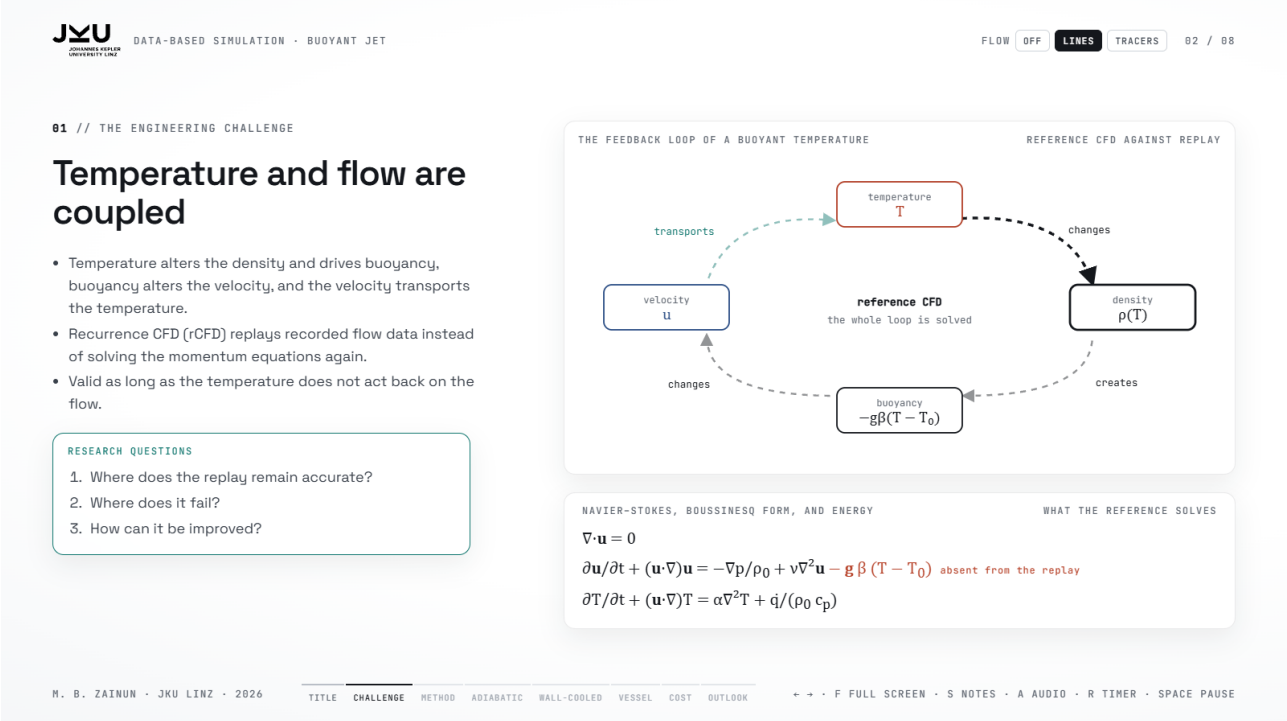
Terima kasih → bersama supervisor, sebut judul → pertanyaan: bisakah aliran terekam dipakai ulang bila suhu mengubah aliran?

Naskah. Thank you for the opportunity to present my work. / I would like to present the work I did with my supervisor, Professor Stefan Pirker: Data-based Numerical Simulation of Buoyant Jet Flow. / The question is simple. Can we reuse a recorded flow, even when the temperature itself changes the flow?

Kata kunci

1. Thanks
2. With supervisor Prof. Pirker, thesis title
3. Question: reuse a recorded flow when T changes the flow

Slide 2 – The engineering challenge *1:00*



Alur cerita

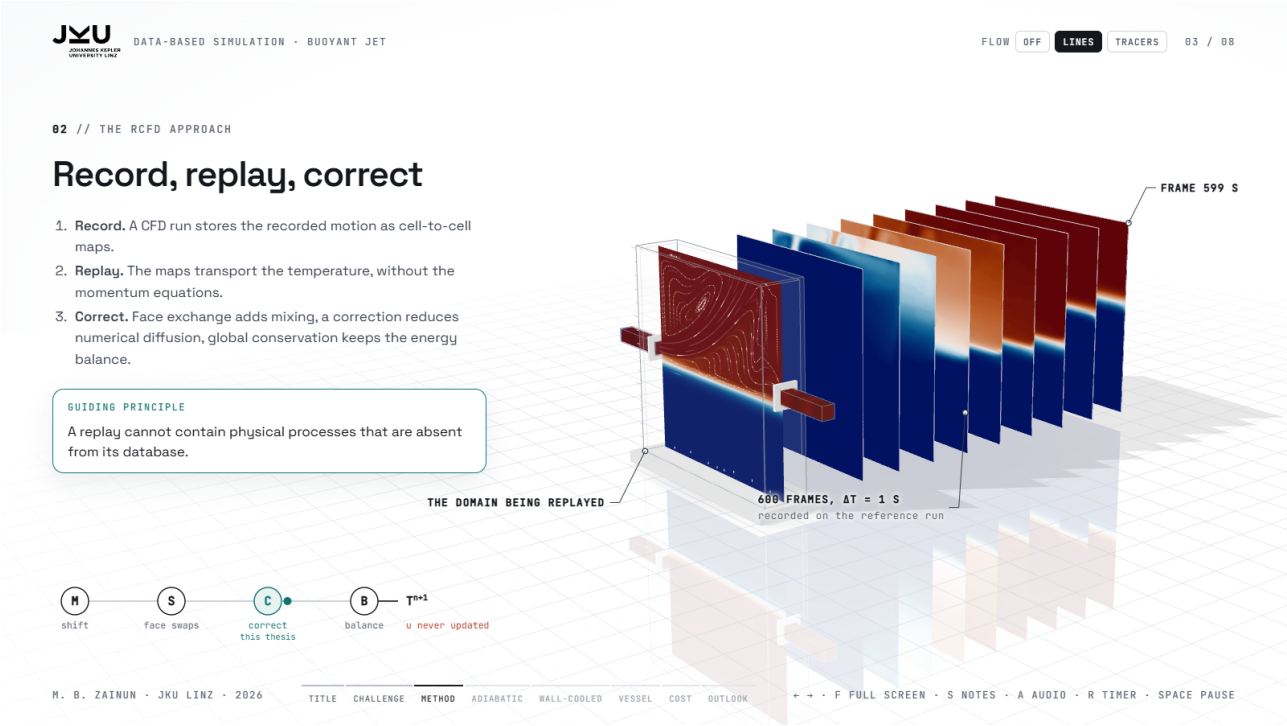
Suhu dan aliran terkopel → air hangat ringan, naik → suku buoyancy menutup loop → rCFD pakai ulang aliran terekam, cocok untuk passive scalar → kasus saya non-passive → tiga pertanyaan.

Naskah. In conventional CFD, temperature and flow are coupled. / Warm water is lighter, so it rises. This buoyancy changes the velocity, and the velocity carries the temperature again. The buoyancy term in the momentum equation closes this loop. / Recurrence CFD, or rCFD, skips the momentum equations and reuses a recorded flow. This is fast, and it works well when the temperature does not act back on the flow, so when it is a passive scalar. In my cases, it is not passive. The temperature drives the flow. / So I asked three questions. Where does rCFD work? Where does it fail? And how can we improve it?

Kata kunci

1. Warm is light, rises → velocity → T
2. Buoyancy term closes the loop
3. rCFD: recorded flow, fine for a passive scalar
4. My cases: non-passive, T drives the flow
5. Three questions: works, fails, improve

Slide 3 – The rCFD method *0:50*



Alur cerita

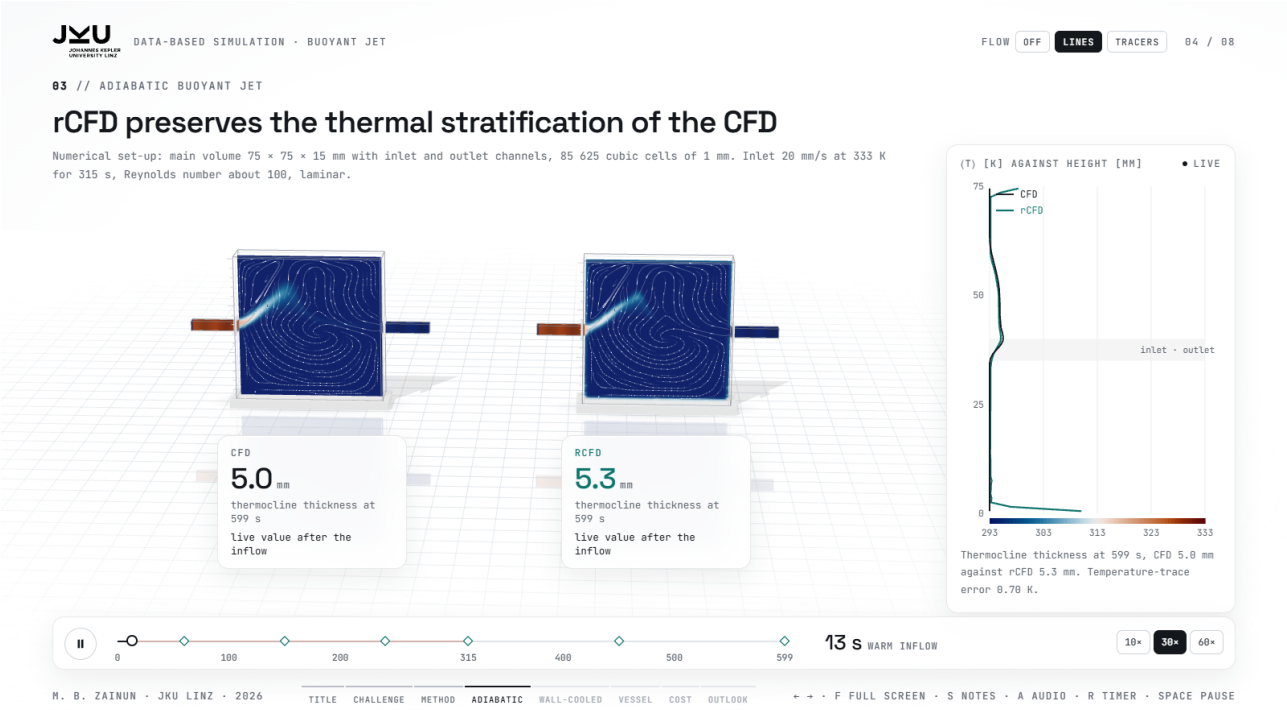
Rekam gerak antarsel (cell-to-cell maps) → replay tanpa persamaan momentum → face exchange dan global conservation → langkah saya: koreksi, lapisan tajam, energi tetap.

Naskah. Let me briefly explain the method. / First, we run a normal CFD simulation and record how the fluid moves between cells. This recorded motion is stored as cell-to-cell maps. / During the replay, these maps transport the temperature without solving the momentum equations again. / But the recorded maps do not capture all of the mixing. So the method adds face exchange between neighbouring cells, and global conservation then corrects the total energy balance. / In my work, I add one more step, a correction. It sharpens the temperature layers that the replay smears out, and it keeps the total energy unchanged.

Kata kunci

1. CFD run → cell-to-cell maps
2. Replay without momentum equations
3. Face exchange, global conservation
4. My step: a correction, sharper layers, same energy

Slide 4 – Numerical set-up and the adiabatic case *1:00*



Alur cerita

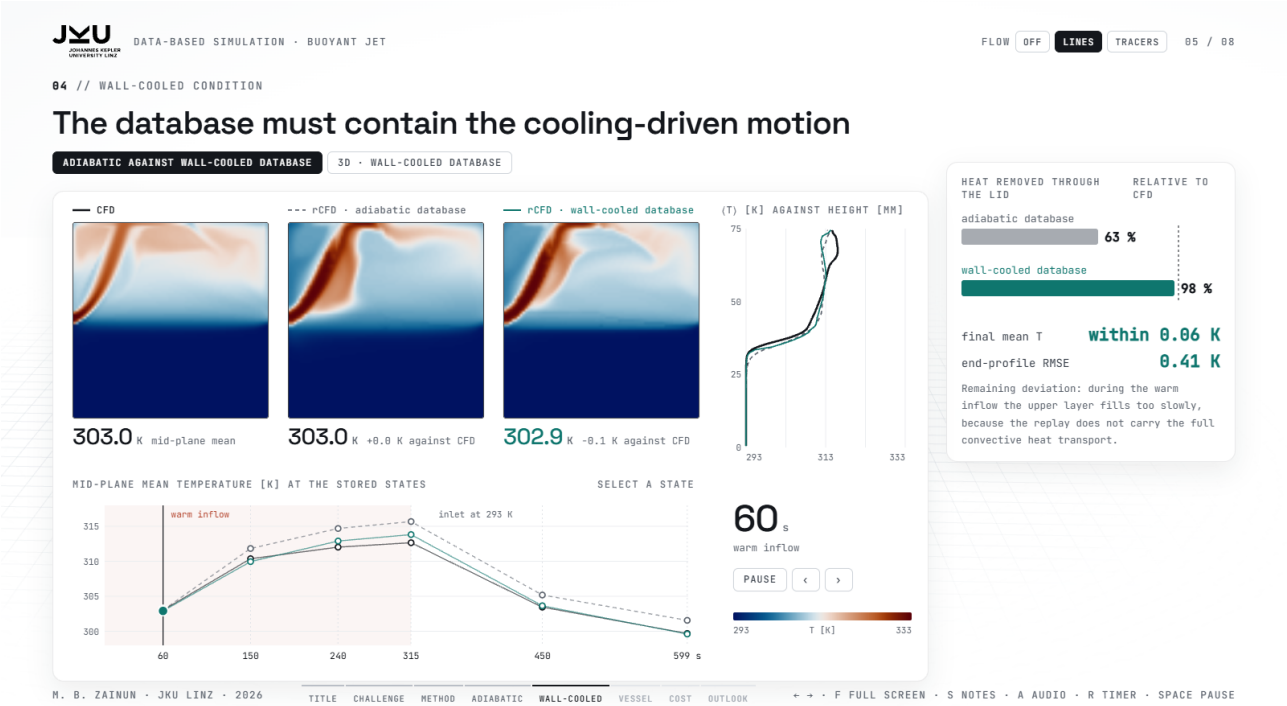
Setup kecil: 75 × 75 × 15 mm + saluran, 85 625 sel 1 mm, Re ≈ 100 laminar → kasus adiabatik → CFD kiri, rCFD kanan → 5,0 mm; tanpa koreksi 15,7 mm; dengan koreksi 5,3 mm, 0,70 K.

Naskah. I started with a small numerical set-up. The main volume is 75 by 75 by 15 millimetres, plus an inlet and an outlet channel, with 85,625 cells of one millimetre. / The inlet Reynolds number is around 100, so the flow is laminar, and we can study the replay errors without turbulence as an additional problem. / The first boundary condition is the adiabatic case. A warm jet enters the cold fluid and forms a warm layer. / On the left you see the CFD, on the right rCFD. In the CFD, the thermocline is 5 millimetres thick at the end of the run. Without the correction, the replay smears this layer to 15.7 millimetres. With the correction, rCFD gives 5.3 millimetres, and the temperature-trace error is 0.70 kelvin. / So in this case, rCFD keeps the thermal stratification of the CFD.

Kata kunci

1. 75 × 75 × 15 mm + channels, 85 625 cells, 1 mm
2. Re ≈ 100, laminar
3. Adiabatic: warm jet, temperature layer
4. 5.0 mm CFD, 15.7 mm without, 5.3 mm with correction, 0.70 K

Slide 5 – Wall-cooled case *1:10*



Alur cerita

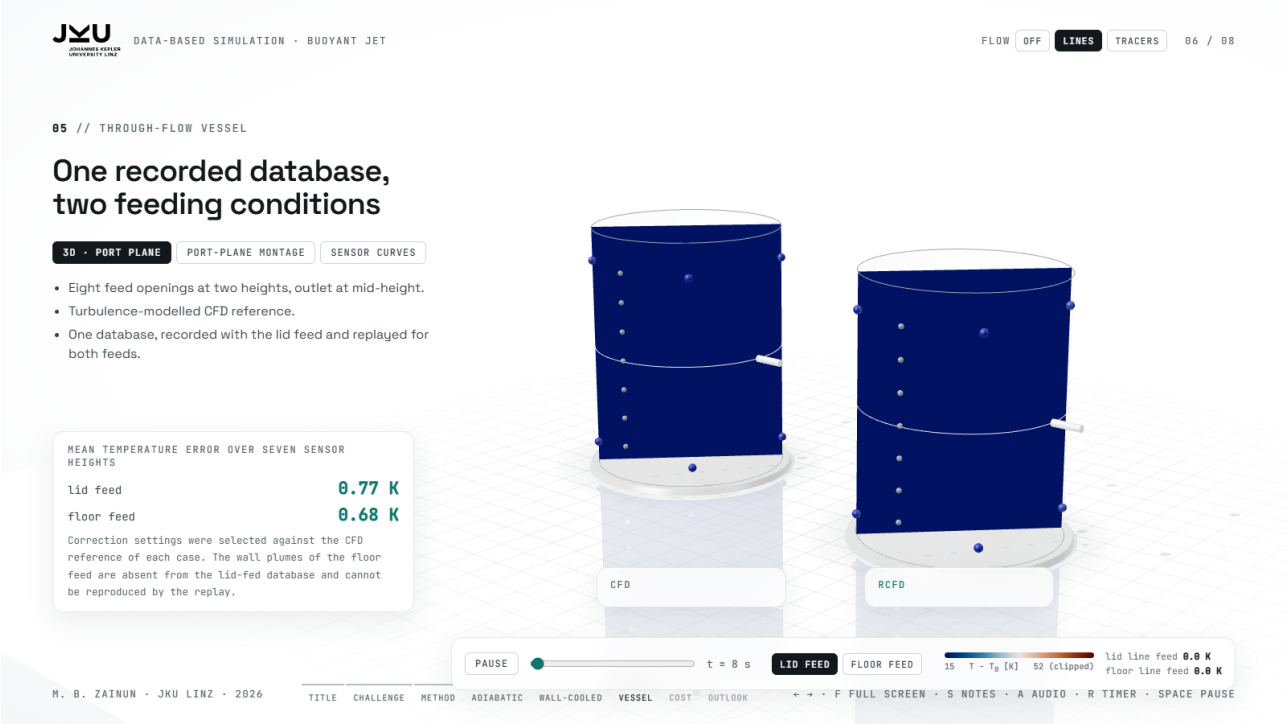
Tutup dingin → cairan tenggelam → database adiabatik 63 %, film dingin di montase → rekam ulang saat didinginkan → 98 %, 0,06 K, 0,41 K → sisa: pengisian lambat.

Naskah. The second is the wall-cooled case. The top wall removes heat. The fluid becomes colder and denser, so it sinks and creates additional motion. / First, I reused the adiabatic database. It removed only about 63 percent of the heat that the CFD removes through the lid. In the montage you can see why. A cold film stays under the lid, and the bulk below stays too warm. The recorded flow never contained the sinking motion, and the correction could not create it. / So I recorded a new database under cooling. Now rCFD removes about 98 percent of the heat. The final mean temperature is within about 0.06 kelvin of the CFD, and the end-profile RMSE is 0.41 kelvin. / One error remains. During the warm inflow, the upper layer fills too slowly, because the replay does not carry all of the convective heat transport.

Kata kunci

1. Top wall removes heat, cold fluid sinks
2. Adiabatic database: 63 %, cold film
3. Wall-cooled database: 98 %, 0.06 K, 0.41 K
4. Remaining: slow filling, convective transport

Slide 6 – Laboratory vessel *0:50*



Alur cerita

Bejana lab → satu database umpan tutup untuk dua umpan → 0,77 dan 0,68 K → setelan dipilih dengan CFD, belum prediksi buta → montase: CFD naik di dinding, rCFD lewat lantai lalu naik di tengah → lapisan benar, jalur salah.

Naskah. Finally, I tested the method in a laboratory vessel with through-flow. / I recorded one database with the lid feed, and replayed it for both feeds. At seven sensor heights, the mean temperature error is 0.77 kelvin for the lid feed and 0.68 kelvin for the floor feed. / The correction settings were chosen with the CFD of each case, so this is not yet a blind prediction. / The montage shows the limit. In the CFD, the warm floor feed rises along the wall. In rCFD, it follows the recorded path, along the floor and up the centre. / The curves still agree, because the sensors give one temperature per height, and the correction keeps warm water above cold water. So rCFD gets the layering right, but not the path.

Kata kunci

1. Lab vessel, one lid-fed database for both feeds
2. 0.77 K and 0.68 K
3. Settings chosen with the CFD, not blind
4. Montage: CFD along the wall, rCFD along the floor and up the centre
5. Layering right, path wrong

Slide 7 – Computational performance *0:35*



Alur cerita

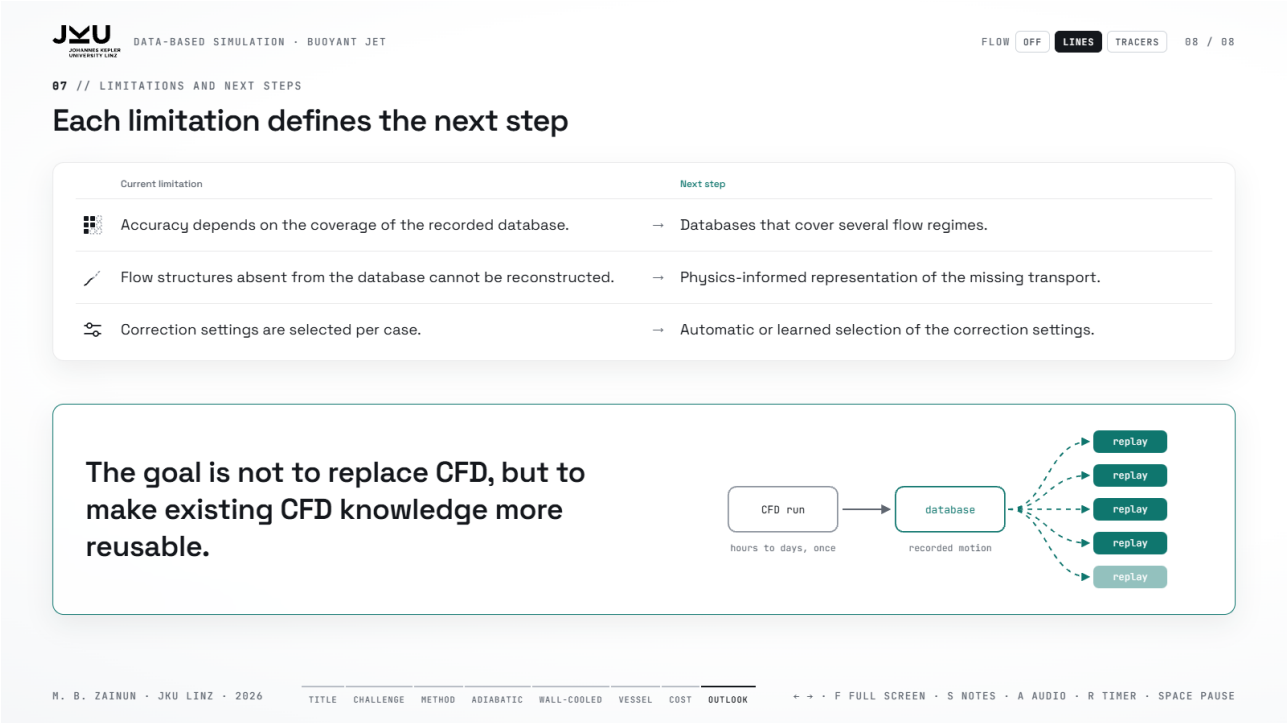
Setelah database ada → sekitar 300 kali, bejana hari jadi menit → biaya database tidak termasuk → satu run CFD untuk banyak replay.

Naskah. Once the database exists, rCFD is much faster. / For the adiabatic case, the CFD took about 569 minutes, and the replay about 1.9 minutes. That is around 300 times faster. For the vessel, the CFD took several days, and the replay less than 9 minutes. / These numbers exclude the cost of generating the database. So one CFD run can feed many replays, for example to test different operating conditions, as long as the database covers them.

Kata kunci

1. 569 → 1.9 min, ≈300×
2. Vessel: days → under 9 min
3. Database cost excluded, one CFD run feeds many replays

Slide 8 – Limitations and next steps *0:45*



Alur cerita

Tiga temuan → batasan → langkah berikutnya → kalimat penutup, terima kasih.

Naskah. To summarise. In the adiabatic jet, rCFD keeps the thermocline of the CFD. Under cooling, the database must contain the cooling-driven motion. And in the vessel, one database serves both feeds. / The limits are clear. The results depend on the database, missing flow structures cannot be reconstructed, and the correction settings are still case-dependent. / My next steps would be better database coverage, automatic correction settings, and physics-informed or learning-based corrections. / The goal is not to replace CFD, but to make existing CFD knowledge more reusable. Thank you.

Kata kunci

1. Three findings
2. Limits
3. Next steps
4. Closing line, thank you

Angka penting

Slide	Parameter	Angka
4	Tebal termoklin, 599 s	CFD 5.0 mm, replay tanpa koreksi 15.7 mm, rCFD dengan koreksi 5.3 mm ($s = 0.40$)
4	Temperature-trace error	0.70 K ($s = 0.40$)
5	Panas lewat tutup	database adiabatik 63 %, database wall-cooled 98 % dari CFD
5	Akhir run	suhu rata-rata dalam 0.06 K, end-profile RMSE 0.41 K
6	Galat tujuh ketinggian	0.77 K (umpan tutup), 0.68 K (umpan dasar)
7	Waktu komputasi	569 → 1.9 min (≈300×), 554 → 2.1 min (≈260×)
7	Bejana	7.4 hari → 8.5 min (≈1250×), 4.7 hari → 8.6 min (≈790×)

Catatan untuk diskusi

- Jangan mengklaim apa pun tentang proses mereka sebelum mendengar dari mereka. Tanyakan proses, data sensor dan skala waktu.
- Kalau ditanya passive atau non-passive: passive scalar tidak memengaruhi aliran, jadi aliran terekam tetap berlaku. Di kasus tesis suhu mengubah densitas dan menggerakkan aliran (non-passive), jadi rekaman hanya berlaku sejauh gerak itu sudah ada di database.
- Kalau ditanya biaya: rekaman tetap butuh satu run CFD penuh, keuntungan datang dari pemakaian ulang.
- Kalau ditanya slide 4: tebal termoklin tergantung kekuatan koreksi s. Di tesis $s = 0.40$ memberi 5.3 mm dan $s = 0.50$ memberi 4.0 mm (Tabel A3). Slide memakai $s = 0.40$, paling dekat ke CFD, dan animasinya juga run $s = 0.40$.
- Kalau ditanya bejana: setelan koreksi dipilih dengan CFD referensi tiap kasus, jadi bukan prediksi buta. Wall plume umpan dasar tidak ada di database.
- Kalau ditanya kenapa kurva cocok tapi medan beda (umpan dasar): kurva = satu suhu per ketinggian, dan koreksi memindahkan panas antar lapisan horizontal sampai kolom stabil, jadi profil ketinggian ikut CFD. Posisi horizontal ikut gerak terekam: di rekaman umpan tutup, jet dasar netral dan naik di sumbu, di CFD umpan dasar air hangat naik di dinding (tesis Bab 7). Ringkasnya: lapisan benar, jalur salah.
- Kalau ditanya pengukuran: perbandingan dengan termokopel rig ada di tesis (Bab 7, nudged runs), tetapi tidak dibawa ke talk ini. Kurva di slide 6 hanya CFD dan rCFD.
- Kalau ditanya sisa galat wall-cooled: pengisian lapisan atas terlambat saat inflow hangat. Uji flux konvektif terekam di tesis hanya diagnostik, bukan bagian metode.
- Kalau ditanya angka 1250×: waktu CFD bejana termasuk render tiap langkah dan laptop yang dipakai bersama run lain (tesis Bab 8), jadi rasio bejana adalah batas atas.
- Animasi: wall-cooled memakai render run $s = 0.475$ (1 Agustus), selisih medannya dengan run tesis 0.25–0.52 K RMS. Angka di slide dari run tesis.
- Semua angka di deck ini ada di tesis (versi 7 November 2026).